

CSMVessel0000000002 V2.0

Project Noah’s Aegis: Client & Employee Cabin Resilience — Induction and Vibration Mitigation

Version 2.0 | June 2026

Changes V1.0 to V2.0

- Aegis Embark vehicle interior products (CSMFAB024-038) directly applied to cabin design
- PVDF-TrFE bio-feedback in cabin bed/mattress: heart rate + respiration monitoring
- Stellar-Tint window film (CSMFAB034): circadian management for passengers crossing time zones
- MXene cabin wall panel (SE=92 dB): replaces V1.0 copper mesh proposal
- ISO 6954:2024 (marine vibration human comfort): updated limits integrated

1. Cabin Vibration Sources V2.0

1.1 Ship-Specific Vibration Profile

Cruise ships generate vibration from: - Main diesel-electric propulsion (typical: 6-cylinder, 120 RPM diesel -> 2 Hz fundamental) - Propeller blade passage frequency (e.g., 6-blade propeller at 120 RPM: 12 Hz) - Auxiliary machinery (pumps, fans: 25-200 Hz range) - Wave encounter (0.05-0.5 Hz; kinetosis-inducing for susceptible passengers)

ISO 6954:2024 passenger comfort limits: | Frequency Band | Comfort Limit (rms) | |—|—| | 0.5-1 Hz | 7.5 mm/s | | 1-2 Hz | 5.0 mm/s | | 2-80 Hz | 2.8 mm/s | | >80 Hz | 0.5 mm/s |

Many cruise ships exceed these limits at propeller blade frequency during sea state 4-5.

1.2 Aegis Embark Applied to Cabins (Cross-Reference CSMFAB Series)

The Aegis Embark vehicle interior product line directly transfers to cabin applications:

CSMFAB Product	Vehicle Application	Cabin Application
031 Mag-Float Seat Rail	Seat vibration isolation	Bed frame vibration isolation
030 Cervical-Guard	Headrest vibration	Pillow + mattress head zone
032 Silence-Block	Door panel acoustic	Cabin wall acoustic insert
028 Cloud-Nest	Child seat base	Infant cot base
034 Stellar-Tint	EV window film	Porthole window circadian film
MAG Bio-Sync	Vehicle seat cushion	Cabin mattress topper

Cabin Aegis Package cost estimate: - Vibration isolation bed frame (custom ship size): \$2,800/cabin - MXene wall panel (SE=92 dB, 2 panels/cabin): \$1,200/cabin - Stellar-Tint porthole film: \$380/cabin - Bio-Sync mattress topper: \$620/cabin **Total per cabin: \$5,000**

2. Schumann Resonance in Cabins V2.0

The Schumann resonance (7.83 Hz) is NOT present inside a steel ship hull — the conductive hull forms a Faraday cage that attenuates ELF frequencies.

This is a known health consideration for long-duration maritime voyages: - Disruption of SR exposure associated with circadian dysregulation (Gerke et al., 2024) - Passengers and crew on 7-day+ voyages experience SR deprivation

V2.0 Solution: KNbO3-BaTiO3 Schumann generators in cabins: Same principle as CSMFAB000000000013 and CSMFAB000000000030: PEMF at 7.83 Hz, 0.5-1 µT amplitude. Power: 0.12 W per cabin, harvested

from ship's power infrastructure. Approval path: IMO MSC circular for passenger wellbeing devices (non-medical).

End of CSMVessel0000000002 V2.0 | Carrington Storm Motors